

MediFlow

Comparing two synthetic healthcare case extracts

CareFlow North and MediTrack East

Analysis report

Analysis run: 15 Apr 2026, 15:33

This report keeps the exploratory checks visible, but all current numbers and tables come from the analysis pipeline.

Purpose

This report follows the same structure as the earlier case work that inspired MediFlow. The aim is simple: check what can be compared across two operational systems, show where the data is strong or weak, and build a repeatable workflow that turns those checks into current figures and tables.

The data in this project is synthetic. The workflow is not. The same kinds of problems are kept on purpose: mixed encodings, mixed date formats, duplicate case IDs, uneven month coverage, and fields that look comparable but should still be treated with care.

Questions this report answers

1. Can the two systems be compared at all, and at what grain?
2. Which source is easier to trust for direct KPI use?
3. What do we need to clean before the numbers are safe to reuse?
4. Which categories look most standardised and highest-volume?
5. What extra patterns show up once the cleaning is done?
6. What should be fixed now, and what should be built next?

Data Basis And Method

The two datasets

The project compares one cleaner system and one messier one.

	CareFlow North	MediTrack East
Raw rows	26,179	17,262
Rows in the 2022 comparison window	26,179	17,165
Date style	One stable YYYYMMDD pattern	Mixed slash and dash timestamps
Month coverage	11/12 months	12/12 months
Main strength	Cleaner process fields	Broader category detail
Main weakness	December registrations are missing	Dates and case IDs need more cleanup

The comparison window is case registrations in 2022. That is why CareFlow stays at 26,179 rows after cleaning, while MediTrack comes down from 17,262 raw rows to 17,165 case-level 2022 rows.

Method

1. Read the files safely, including encoding checks.
2. Keep date columns as text first and inspect the raw formats.
3. Parse dates only after the actual source patterns are clear.
4. Separate exact row duplicates from duplicate case IDs.
5. Mark treatment-time metrics as robust or exploratory before plotting them.
6. Generate figures, tables, and export files from the same pipeline run.

What can be compared?

The systems do *not* support a row-by-row join. They do support a useful case-level comparison because the same business ideas appear in both sources.

Shared idea	CareFlow North	MediTrack East	Use in this report
Case reference	CARE_REF	RefNo	Same case grain idea, but not a shared key
Registration date	DT_REG	RegisteredAt	Safe for year and month comparisons
Assignment date	DT_ALLOC	AssignedOn	Safe for order checks
Cancelled date	DT_ABORT	CancelledOn	Safe for cancellation-rate comparison
Pathway start	DT_OPEN	StartPathway	Useful context, not the main KPI basis
Finished date idea	max(DT_END, DT_CLOSE)	FinishedAt	Comparable in concept, but weaker on the MediTrack side
Geography proxy	ZIP_PAT	PostalArea	Used for map views only
Category field	TRACK	CareGroup	Used for category-pattern analysis

Bottom line The project compares the two systems at case level and on shared concepts. It does not pretend they are the same source or that every KPI is equally strong in both.

Data Quality

The two systems need different kinds of help before they can sit in the same report. CareFlow is easier to parse and easier to trust for direct KPI work. MediTrack gives richer category patterns, but the pipeline has to do more validation first.

Comparative overview

Dimension	CareFlow North	MediTrack East
Date format	OK: single YYYYMMDD format	Caution: mixed slash/dash timestamps (50.1% slash, 49.9% dash)
Parse rate	OK: 100.0%	OK: 100.0%
Reg <= assign	OK: 99.8%	Caution: 82.3%
Finish before reg	OK: 0.156%	Fail: 46.0%
Treatment-time KPI	OK: median 48 d	Exploratory: median 57 d
Temporal coverage	Caution: 11/12 months	OK: 12/12 months
Duplicates	OK: no duplicate case rows	Fail: 81 exact rows and 16 extra case rows required collapse

Data quality scorecard

Dimension	CareFlow North
Date format	OK: single YYYYMMDD format
Parse rate	OK: 100.0%
Reg <= assign	OK: 99.8%
Finish before reg	OK: 0.156%
Treatment-time KPI	OK: median 48 d
Temporal coverage	Caution: 11/12 months
Duplicates	OK: no duplicate case rows

Dimension	MediTrack East
Date format	Caution: mixed slash/dash timestamps (50.1% slash, 49.9% dash)
Parse rate	OK: 100.0%
Reg <= assign	Caution: 82.3%
Finish before reg	Fail: 46.0%
Treatment-time KPI	Exploratory: median 57 d
Temporal coverage	OK: 12/12 months
Duplicates	Fail: 81 exact rows and 16 extra case rows required collapse

Scorecard values are generated from the current analysis run, not re-entered by hand.

Figure 1: Data-quality scorecard figure generated from the current run.

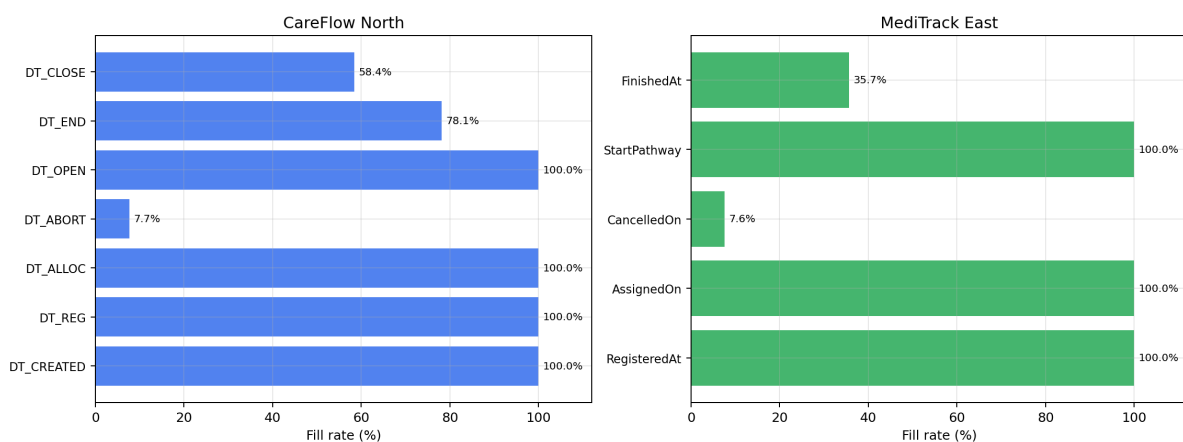


Figure 2: Date-field completeness. This is useful because a clean format still does not guarantee a complete process trace.

CareFlow: more consistent, but not complete

CareFlow is the easier source to work with. It keeps one date style, has almost no finish-before-registration cases (0.156%), and supports a direct finished-case rate of 78.1%.

The main weakness is scope, not parsing. CareFlow only covers 11/12 months, so the year view is intentionally incomplete. It also keeps `DT_CREATED` as an older administrative timestamp, which is why the clean system still shows pre-2022 dates in the date-span checks.

MediTrack: fuller year, messier dates

MediTrack covers 12/12 months, but it needs more cleanup before the KPIs make sense. The `RegisteredAt` field splits almost exactly in half between slash and dash styles (50.1% slash / 49.9% dash). The pipeline also removes 81 exact duplicate rows and collapses 16 extra case rows before the comparison reaches case level.

The important warning is date order. 46.0% of finished cases still fall before registration, so treatment time stays an exploratory check on the MediTrack side even after parsing succeeds.

Date Cleaning And Excel Formulas

One reason to keep the exploration visible is that the first useful output in a real project is often not a finished report. It is a small practical fix, like helping a department turn exported date strings into usable dates in Excel.

What we checked first

- We tested candidate file encodings to find an import setup that preserved the raw values.
- We read date columns as text first to find the real source formats before parsing them.
- We measured the slash and dash populations to find how mixed the messy date fields were.
- We separated exact duplicate rows from duplicate case IDs to find the true case grain.
- We marked risky metrics as exploratory to find which comparisons were safe to treat as headline results.
- We generated report facts from the script to keep the written outputs aligned with the current run.

Excel formulas kept as working notes

A practical part of the original exploration was helping a department turn exported date strings into usable Excel dates before any KPI work could start.

Pattern		Excel formula used in the exploration
Clean export (YYYYMMDD)		<code>=IF(A2=0,"",DATE(LEFT(TEXT(A2,"00000000"),4),MID(TEXT(A2,"00000000"),5,2),RIGHT(TEXT(A2,"00000000"),2)))</code>
Mixed export (slash / dash timestamps)		<code>=IF(A2="", "", IF(ISNUMBER(SEARCH("/",A2)),DATE(MID(A2,FIND("/",A2),FIND("/",A2)+1)+1,4),LEFT(A2,FIND("/",A2)-1),MID(A2,FIND("/",A2)+1,FIND("/",A2,FIND("/",A2)+1)-FIND("/",A2)-1)),DATE(RIGHT(LEFT(A2,10),4),MID(A2,4,2),LEFT(A2,2))))</code>

These formulas are kept as documentation of the exploration. The Python pipeline now mirrors the same logic, so Excel is not the source of truth for the final report.

Dates outside the 2022 window

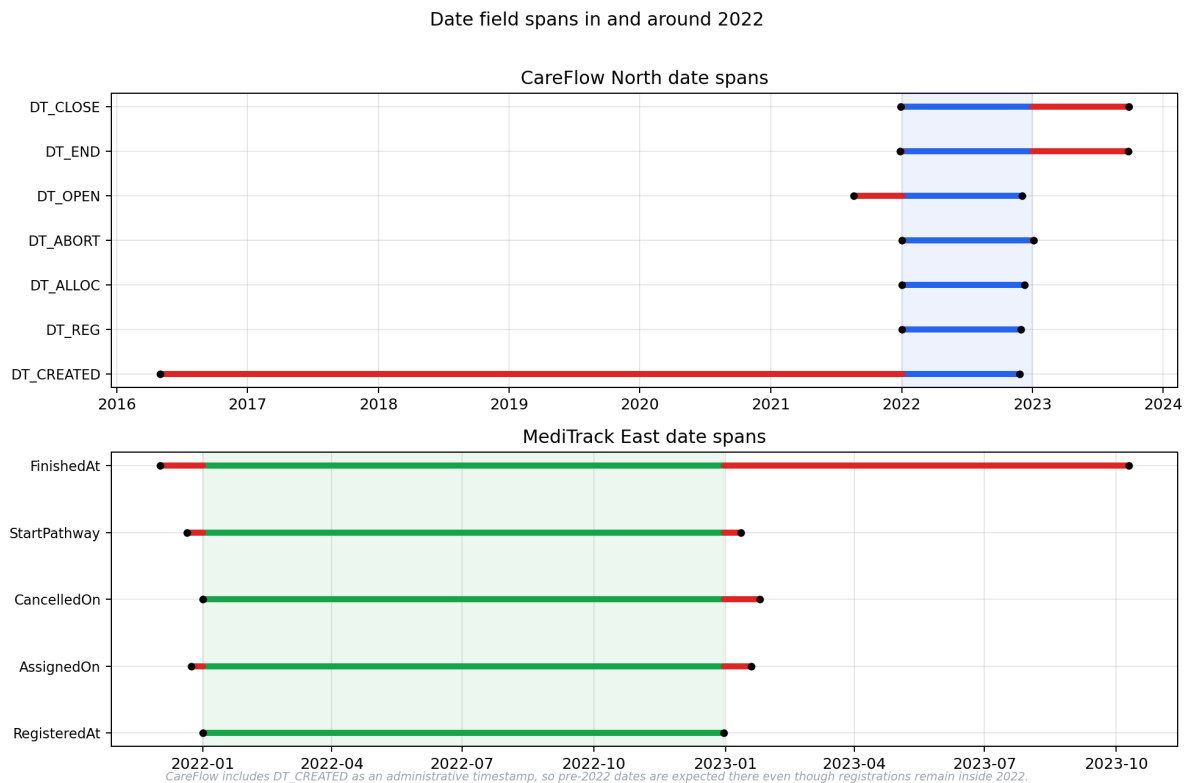


Figure 3: Date-field spans in and around 2022. CareFlow includes older admin timestamps, while MediTrack shows the effect of mixed process timing and later activity.

KPI Overview

After cleaning, the project can give a stable case-level comparison. The important difference is that not every KPI should be used with the same confidence.

Metric	CareFlow North	MediTrack East	Note
Registered cases (2022)	26,179	17,165	Case-level comparison
Finished cases	20,456	6,123	Completed vs Done
Finished case rate	78.1%	35.7%	Same concept
Open rate	14.1%	56.8%	MediTrack groups Open, Pending, and In Progress
Cancelled rate	7.7%	7.6%	Comparable
Median treatment time	48 days	57 days	MediTrack exploratory
Finish before registration	0.156%	46.0%	Date-order check

[AUTO-GENERATED INTERPRETATION] CareFlow supports a more direct KPI read in this run: 78.1% finished, 14.1% still open, and almost no finish-before-registration cases (0.156%). MediTrack is still useful for case counts and status mix, but 46.0% of finished cases fall before registration, so the 57 days treatment-time figure should stay a rough check.

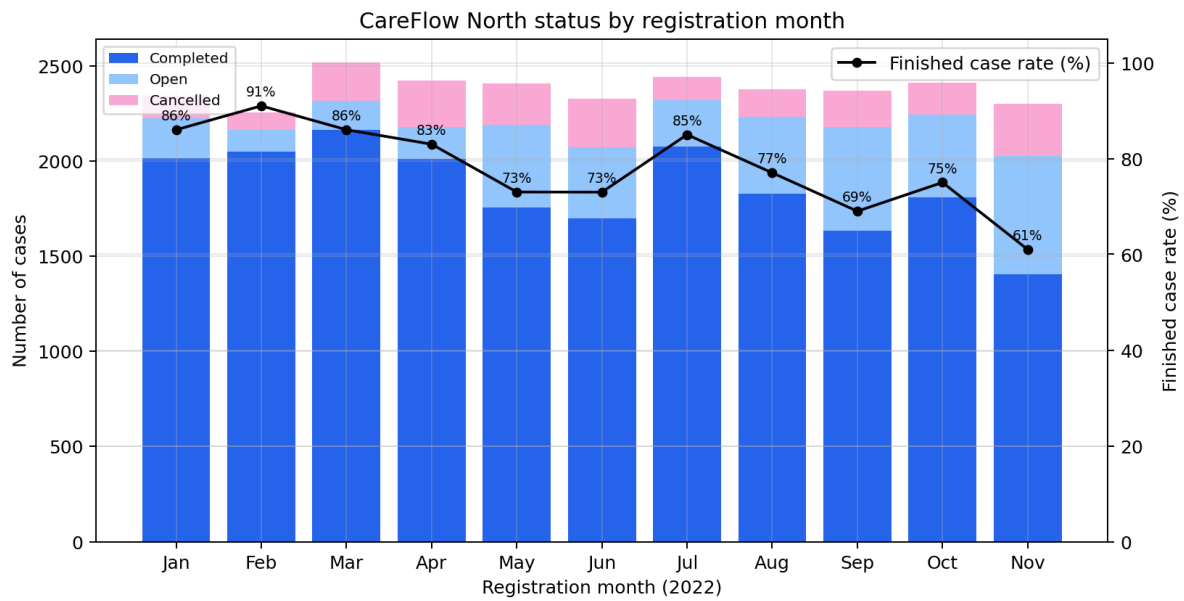


Figure 4: CareFlow monthly status. The clean system still keeps a real backlog and a non-flat cancellation pattern.

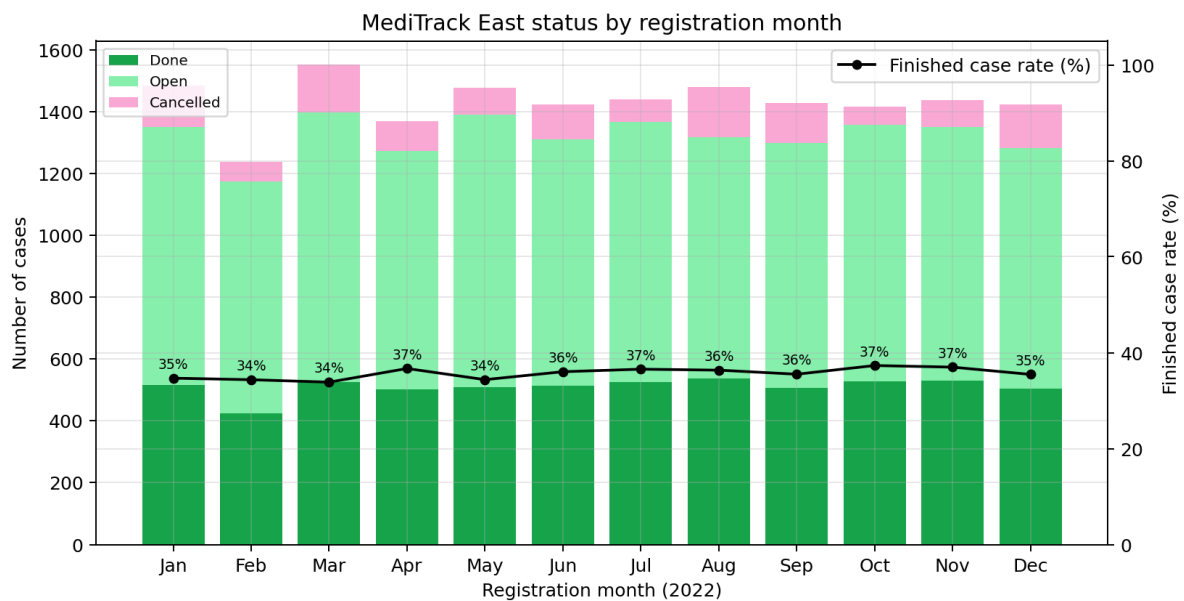


Figure 5: MediTrack monthly status. Open, pending, and in-progress states are grouped into one open view here so the comparison stays readable.

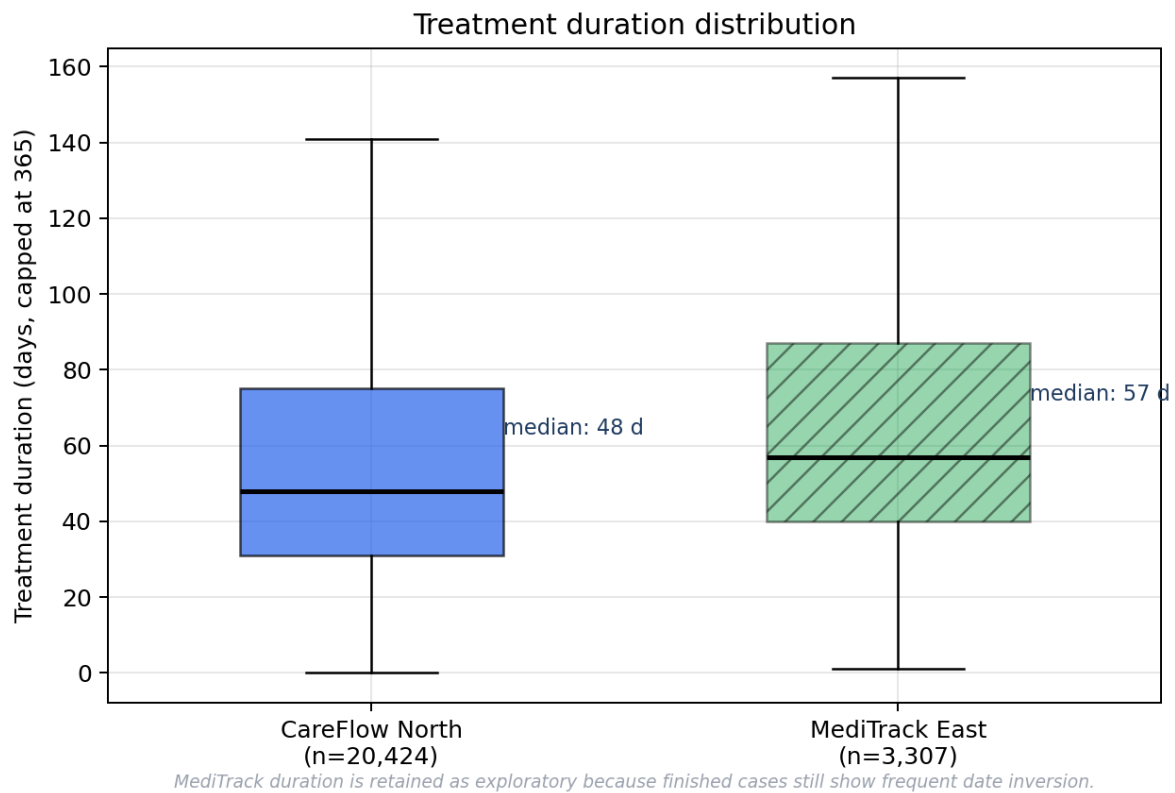


Figure 6: Treatment-time distribution for finished cases. CareFlow is suitable for headline use; MediTrack is still shown as an exploratory timing check.

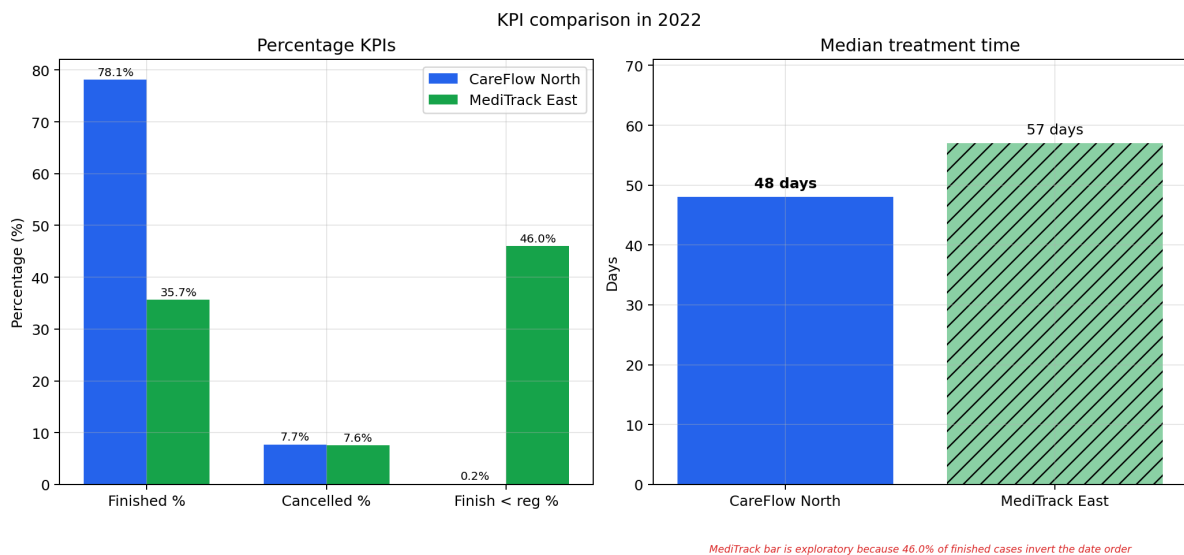


Figure 7: Compact KPI comparison. The treatment-time panel keeps the MediTrack warning visible.

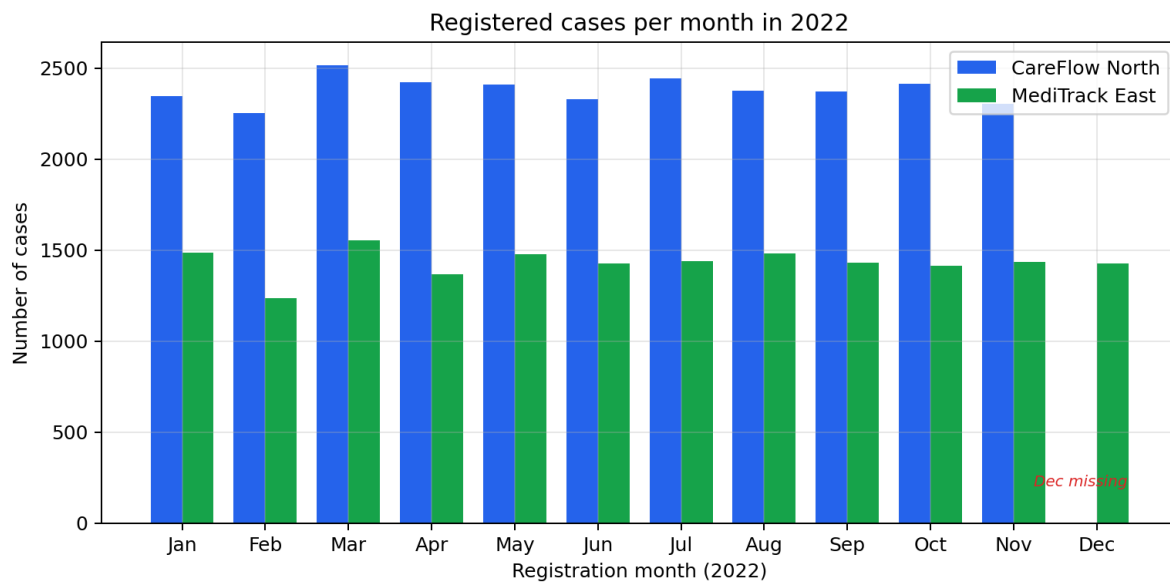


Figure 8: Monthly registration volume. This gives a simple read on the case load across the 2022 study window.

Recommendations

The old case ended with practical recommendations. That structure still works well here.

Fix now

#	Recommendation	Why it matters
1	Standardise date exports	MediTrack still mixes slash and dash timestamps in the same field. One clear export pattern would remove a whole layer of cleanup.
2	Define a complete reporting window	CareFlow is clean enough for KPI work, but the missing December cases mean the year view is still incomplete.
3	Keep case grain explicit	The MediTrack comparison only becomes stable after exact row deduplication and case-level collapse on RefNo.

Build next

#	Recommendation	Why it matters
1	Publish field definitions with each extract	The report is stronger when field meaning is explicit, especially for finished-date logic.
2	Keep robust and exploratory KPIs separate	That makes it clear which numbers are safe headlines and which are still checks or prompts for follow-up.
3	Automate the intake checks	Encoding checks, date parsing, duplicate handling, month coverage, and report refresh are already scriptable and should stay together.

Category Patterns And Geography

MediTrack category pattern

The old presentation used category analysis mainly on the messier system, because that was where the readable category labels lived. The same logic still applies here.

CareGroup	Share	Completed	Cancelled
Kort Sag	28.0%	37.3%	6.0%
Pakke A	22.1%	34.8%	8.4%
Pakke B	17.5%	32.8%	10.4%
Forlob Y	12.8%	38.8%	4.5%
Revurdering	10.9%	30.6%	12.6%
Forlob X	8.7%	40.2%	3.0%

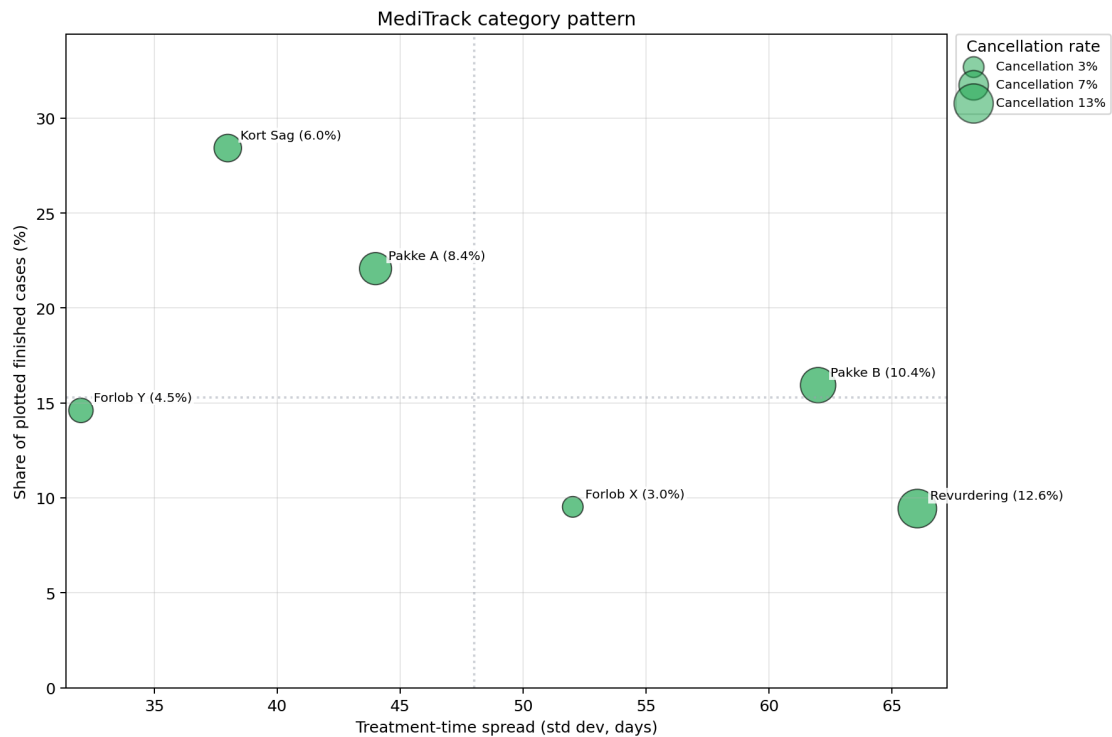


Figure 9: MediTrack category pattern. The y-axis now uses relative share, which makes more sense than raw finished counts after cleanup.

Added view: CareFlow and combined patterns

MediFlow adds one thing the original case did not have: a matching category view for the cleaner system and a combined view across both systems. That makes it easier to see that category-level spread matters more than a simple clean-versus-messy split.

TRACK	Share	Completed	Cancelled
HOME_MON	22.9%	78.3%	7.5%
SURG_PRE	19.9%	79.7%	6.2%
ACUTE	14.9%	75.5%	10.4%
REHAB	13.8%	76.8%	9.1%
OUTP_FUP	10.7%	81.2%	4.6%
SURG_POST	9.5%	73.3%	12.6%
DIAG	8.4%	82.5%	3.4%

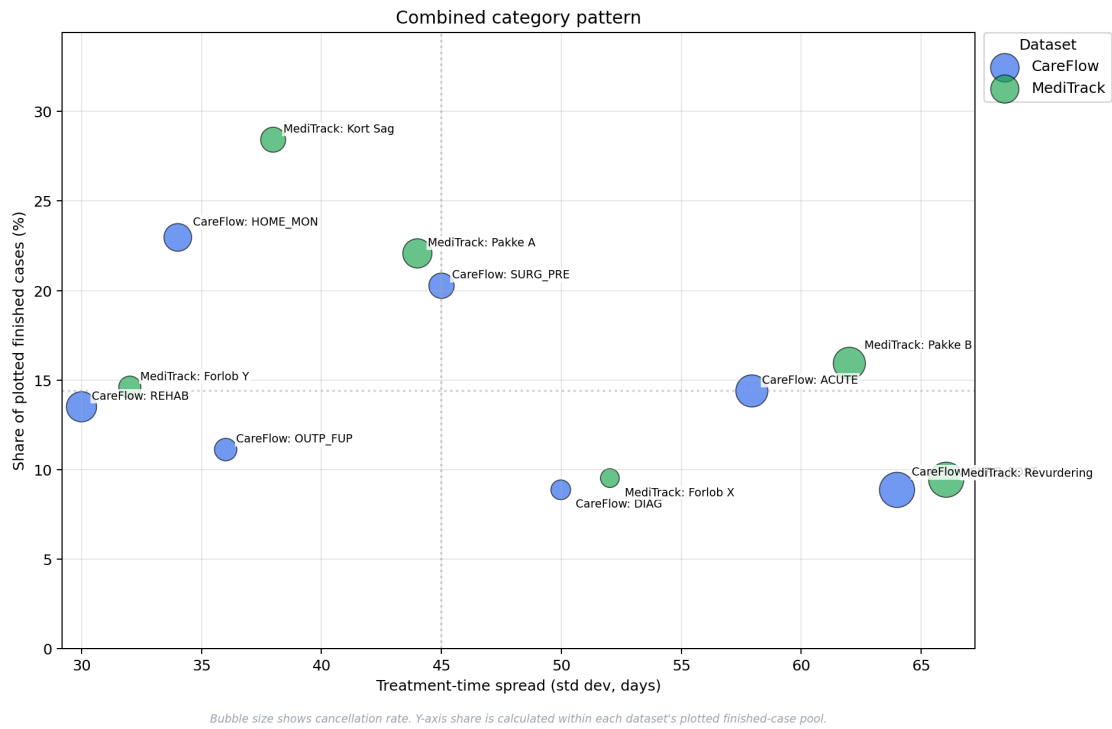


Figure 10: Combined category pattern. This extra view was added in MediFlow to compare category spread and share across the two systems on the same chart.

Synthetic postcode pools on real geography

CareFlow pool	Share	MediTrack pool	Share
Hovedstaden	93.9%	Hovedstaden	3.5%
Sjaelland	5.0%	Sjaelland	94.6%
Other	1.1%	Other	1.9%

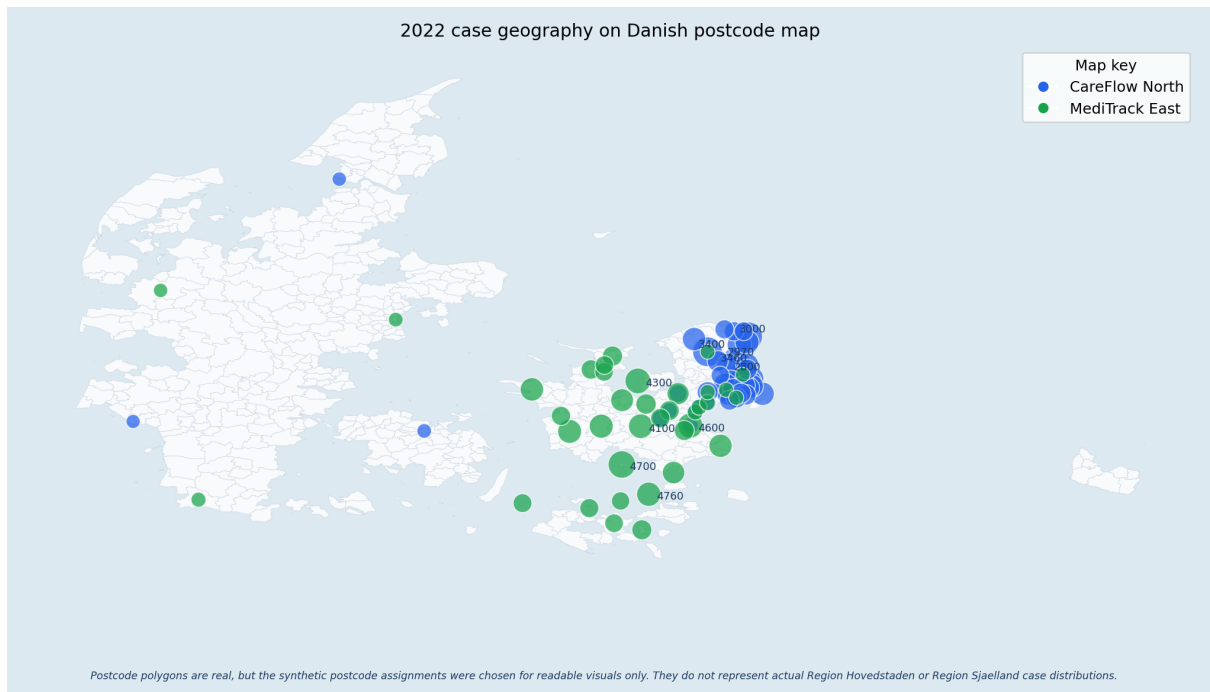


Figure 11: Synthetic postcode assignments on real Danish postcode geometry. The shapes are real; the postcode choices are only there to make the visual comparison readable.

[AUTO-GENERATED INTERPRETATION] CareFlow mostly uses Hovedstaden postcodes (93.9%) with a smaller Sjælland share (5.0%). MediTrack mostly uses Sjælland postcodes (94.6%) with some Hovedstaden spillover (3.5%). The postcode pools are synthetic and were chosen for readable visual separation, not to describe real regional activity.

Conclusion

This run supports three plain conclusions:

1. The two systems can be compared at case level and on shared concepts, but not by joining rows across systems.
2. CareFlow is the stronger source for direct KPI reporting, while MediTrack is still better treated as a richer but messier source that needs more caution.
3. The biggest immediate value is in the data workflow itself: stable import, stable parsing, clear case grain, and report refresh from one pipeline run.